Curriculum Vitae

CHARLES MICHAEL LEWIS

Address:

School of Computing and Information
Department of Informatics and Networked Systems
University of Pittsburgh, Pittsburgh, PA 15260
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RESEARCH INTERESTS

Human-Robot Interaction, Human Factors, Human Computer Interaction, Visualization, CSCW, Virtual Environments, Software Agents, Human Error

TEACHING INTERESTS

Human Factors, Human Computer Interface, Human Computer Interaction, Decision Making/Decision Support, Interactive Systems

EDUCATION

Ph.D. in Psychology (engineering psychology), Georgia Institute of Technology, 1986, Thesis: Identification of Rule-Based Models. Advisor: W.B. Rouse

M.S. in Psychology (engineering psychology), Georgia Institute of Technology, 1982, Thesis: Modeling from Video Display in Repair Performance. Advisor: E.J. Rinalducci

B.A. in Experimental Psychology, University of California, Santa Barbara, 1973

PROFESSIONAL EMPLOYMENT and EXPERIENCE

Professor 2006-present School of Computing and Information, Department of Informatics and Networked Systems with joint appointment in Intelligent Systems Program, University of Pittsburgh

Associate Professor 1993-2006 Department of Information Science and Telecommunications, School of Information Sciences with joint appointment in Intelligent Systems Program, University of Pittsburgh

Assistant Professor 1987-1993 Department of Information Science and Telecommunications, School of Information Sciences, University of Pittsburgh

Senior Human Factors Engineer 1986 Iowa Electric Light & Power Co., Cedar Rapids, IA, Lead HFE on Control Room Design Review of Duane Arnold nuclear power plant

Graduate Research Assistant and Instructor 1984-1986, Center for Human-Machine Systems Research, School of Industrial and Systems Engineering, Georgia Institute of Technology, Atlanta, GA 30332

Associate Scientist 1984-1986, General Physics Corp., 1770 The Exchange NW, Atlanta, GA

Human Factors Specialist 1980-1984, Human Performance Group, Institute of Nuclear Power Operations, 1100 Circle 75 Parkway, Atlanta, GA

Contract and Grant Support to Michael Lewis

- Stefanos Nikolaidis, Katia Sycara and **Michael Lewis,** Exploring Diversity in Multi-Human AI Teams, (April 2024-October 2025), *Defense Advanced Research Projects Agency*

- **Michael Lewis** and Katia Sycara, Testing and Evaluation for Soldier-Device Teaming Compatibility, Vulnerability, and Durability in Emergent Situations: 3.6 Characterization of Human Control of Multiple Unmanned Systems, 4.3 Updating Metrics and Methods for Evaluating Human-Robot Teams (January 2022-July 2025) *DEVCOM Data Analysis Center*

- Katia Sycara and **Michael Lewis**, Strengthening Robust Teamwork in Human Agent Teams Cycle 3 (seedling), (March 2021- February 2022), *Army Research Laboratory*

- Katia Sycara and **Michael Lewis**, Strengthening Robust Teamwork in Human Agent Teams Cycle 2 (seedling), (March 2020- February 2021), *Army Research Laboratory*

- Katia Sycara, **Michael Lewis** and Julie Shah, *Individualized Adaptation in Human Agent Teams* (May 2020-April 2023), *Army Research Laboratory*

- Katia Sycara, **Michael Lewis** and Ryan McFarland, *A Robust and Adaptive Agent that Supports High Performance Teams* (December 2019-December 2023), *Defense Advanced Research Projects Agency*

- **Michael Lewis**, *ISHIS—Interface System for Human Interaction with Swarms* (November 2019-October 2022), *Office of Naval Research*

- Katia Sycara and **Michael Lewis**, *Individualized Adaptation in Human Agent Teams* (seedling), (March 2019- February 2020), *Army Research Laboratory*

- Katia Sycara, Nilanjan Chakraborty, and **Michael Lewis**, *Trustworthy Human Interaction with Robotic Swarms*, (June 2015-June 2020), *Air Force Office of Scientific Research*

- Katia Sycara, Christian Lebiere, Meeko Oishi, and **Michael Lewis**, *Formal Models of Human Control and Interaction with CPS*, *National Science Foundation*, (September 2013-September 2016), *Air Force Office of Scientific Research*.

- Katia Sycara and **Michael Lewis**, *The influence of Cultural Factors on Trust in Automation*, (June 2013-May 2016), Air *Force Office of Scientific Research*.

- Katia Sycara, Paul Scerri, **Michael Lewis**, Michael Goodrich, *Cognitively Compliant Command for Multirobot Teams*, Science of Autonomy, (May 2009-May 2014), *Office of Naval Research*

- Katia Sycara, Baruch Fischhoff, Geoffrey Gordon, **Michael Lewis**, Katherine Tinsley, Robin Dillon, Scott Atran, Robert Axelrod, Charles Grindle, David Traum (May 2008-April 2013), *Cultural Models of Collaboration and Negotiation*, Multidisciplinary University Research Initiative*, Army Research Organization*.

- Katia Sycara, Paul Scerri, Christian Lebiere, **Michael Lewis**, Jonathon How, Mary Cummings, Mark Campbell, and Raja Parasuraman (May 2008-April 2013), *Modeling Synergies in Large Human-Machine Networked Systems*, Multidisciplinary University Research Initiative, *Air Force Office of Scientific Research*.

- Katia Sycara and **Michael Lewis** (December 2006-November 2009) *Large Scale Computation for Human Agent Team Coordination in Dynamic Environments*, *Air Force Office of Scientific Research*.

- Eduardo Salas, Nancy Cooke, **Michael Lewis** and Alex Kirlik (May 2006-April 2011) *Reconfigurable Human-Agent Collaboration: Human Performance in Network-Centric Operations*, Multidisciplinary University Research Initiative, Office *of Naval Research*.

- **Michael Lewis** (April 2005-April 2006) *Extending USARsim to Support MOAST Framework*, *National Institute of Standards and Testing*.

- Katia Sycara and **Michael Lewis** (May 2003-November 2005) Cooperative *Attack Testbed for Aircraft, Munitions, and Ground Vehicles*, Air *Force Research Laboratories AFRL/MN*.

- Katia Sycara, Illah Nourbakhsh and **Michael Lewis** (July 2002-July 2006) *Coordination of Heterogeneous Teams (Humans, Agents, Robots) for Emergency Response*, Information Technology Research Initiative, *National Science Foundation*.

- Katia Sycara and **Michael Lewis** (September 2001-August 2006) *Information Fusion for Command and Control: from Data to Actionable Knowledge and Decision*, Air *Force Office of Scientific Research*

- **Michael Lewis** (January 1999-December 1999) Smooth *Navigation through a Cluttered Virtual Environment*, Fellowship in Advanced Simulation and Training, Supervision for Jeffrey Jacobson, *Link Foundation*

- Katia Sycara and **Michael Lewis** (September 1997-February 2002) *Integrating Intelligent Assistants into Human Teams*, Multidisciplinary University Research Initiative, *Office of Naval Research*.

- Robert Korfhage and **Michael Lewis** (March 1996-February 1999) *Visual Information Retrieval Interfaces*, Interactive Systems Program*, National Science Foundation*.

- **Michael Lewis** (January 1991-December 1993) Ecological *Cognition in Visualization*, Interactive Systems Program, *National Science Foundation*.

Editorial Boards

IEEE Transactions on Systems, Man, and Cybernetics: Part A, now reorganized as *IEEE Transactions on Human Machine Systems* (past associate editor)

Human Factors (Journal of the Human Factors and Ergonomics Society) (past associate editor)

Group Decision and Negotiation: Kluwer Academic Publishers (affiliated with Institute for Operations Research and Management Sciences) (past associate editor)

IEEE International Conference on Robotics and Automation (ICRA)

Autonomous Agents and Multi-Agent Systems: Kluwer Academic Publishers (past associate editor)

Cognition, Technology, and Work: International Journal for the Analysis, Design, and Use of Joint Cognitive Systems: Springer-Verlag (past associate editor)

Reviewer

Grant proposals National Science Foundation, Interactive Systems Program, various programs (1991-present)

IEEE Transactions on Systems, Man, and Cybernetics

IEEE Transactions on Robotics

IEEE Transactions on Reliability

Journal of Visual Languages and Computing

International Journal of Human Computer Studies

Journal of Field Robotics

IEEE International Conference on Systems, Man, and Cybernetics

International Joint Conference on Autonomous Agents and Multiagent Systems

Human Factors and Ergonomics Society Annual Meeting, Computer Systems Technical Group, Cognitive Engineering and Decision Making, and general sessions

ACM/IEEE Conference on Human-Robot Interaction

IEEE International Conference on Robotics and Automation

IEEE/RSJ International Conference on Intelligent RObots and Systems

RoboCup Rescue

RoboCup 2008 team placed 3rd in Virtual Robots Urban Search & Rescue Competition

RoboCup 2007 Winner Virtual Robots Urban Search & Rescue Competition

RoboCup 2006 User Interface Award, Virtual Robots Urban Search & Rescue Competition

American Open 2006 team placed 1st in autonomy

Member of RoboCup Organizing Committee 2004 & 2005

American Open 2005, team placed 1st in mobility, 1st in autonomy and 3rd in rescue competitions

American Open 2004, team placed 3rd in rescue competition

Member

Association for Computing Machinery
Special Interest Group on Computer Human Interaction

Institute of Electrical and Electronics Engineers, Lifetime Member

Computer Society
Systems, Man, and Cybernetics Society

Human Factors and Ergonomics Society
Computer Systems Technical Group
Cognitive Engineering and Decision Making Technical Group

Sigma Xi

Dissertation Chair

Huao Li, August 2025, A Computational Framework for Efficient Human-agent Teamwork, School of Computing and Information, University of Pittsburgh, Postdoctoral Fellow for Engineering Excellence in the Department of Aeronautics and Astronautics, Massachusetts Institute of Technology, Cambridge, Massachusetts, huaol@mit.edu

Saman Amirpour Amraii, April 2018, Human-Data Interaction in Large and High-Dimensional Data, School of Computing and Information, University of Pittsburgh, Co-founder, xSeer, Palo Alto, CA, nebeleh@gmail.com

Phillip Walker, April 2017, Improving Operator Recognition and Prediction of Emergent Swarm Behaviors, School of Information Sciences, University of Pittsburgh, Research Scientist, Soft Information Flow Technologies (SIFT), Minneapolis, MN, pwalker@sift.net

Shih-Yi Chien, April 2017, The Influence of Cultural Factors on Trust in Automation, School of Information Sciences, University of Pittsburgh, Associate Professor, National Chengu University, TW, gsechien@gmail.com

Shafiq Abedin April 2016, *3D Generalization of Brain Model to Visualize and Analyze Neuroanatomical Data*, School of Information Sciences, University of Pittsburgh, Research Scientist, IBM Research-Almaden, sabedin@us.ibm.com

Huadong, Wang, August 2013, *Asynchronous Visualization of Spatiotemporal Information for Multiple Moving Targets*, School of Information Sciences, University of Pittsburgh, First Senior UX Research Manager, Zoox, San Francisco, CA, jwhd82@gmail.com

Jijun Wang, April 2008*, Human Control of Cooperating Robots*, School of Information Sciences, University of Pittsburgh, Software Engineer, Intel Corp, Philadelphia, PA., jijunwang.cn@gmail.com

Jumpol Polvichai, April 2008, *Modeling Team Performance for Coordination Configurations of Large Multi-Agent Teams*, School of Information Sciences, University of Pittsburgh, Associate Professor, King Mungkut's Institute of Technology (Thailand), jumpol@cpe.kmutt.ac.th

Yang Xu, April 2008, *Token-Based Approach for Scalable Team Coordination*, School of Information Sciences, University of Pittsburgh, Professor, University of Electronic Science and Technology, Chengdu, China, xuyang.uestc@me.com

Jeffrey Jacobson, April 2008, *Ancient Architecture in Virtual Reality; Does Immersion Really Aid Learning?* School of Information Sciences, University of Pittsburgh, XR Project Manager at Boston Children's Hospital, jeff@enterprisevr.com

Stephen Hughes August 2005. *Assisted Viewpoint Interactions for 3D Visualizations*, Department of Information Science and Telecommunications, School of Information Sciences, University of Pittsburgh, Associate Professor, Coe College, Cedar Rapids, IA, shughes@coe.edu

Cholyeun Hongpaisanwiwat December 2002. *The Influence of Animated Characters on Comprehension and Attention Performance in a Multimedia Presentation*, Department of Information Science and Telecommunications, School of Information Sciences, University of Pittsburgh, Assistant Professor Thammasat University (Thailand), chhong@cs.tu.ac.th

Terri Lenox May 2000. *Supporting Teamwork Using Software Agents in Human-Agent Teams*, Department of Information Science and Telecommunications, School of Information Sciences, University of Pittsburgh, Professor and Chair of Mathematics and Computer Science and Chair of the faculty, Westminster College, retired, lenoxtl@westminster.edu

Jeffrey Campbell May 2000. *Concurrency Control for Collaborative Diagramming*, Department of Information Science and Telecommunications, School of Information Sciences, University of Pittsburgh, Ecological Data Scientist, USDA Agricultural Research Service, jeffrey.campbell@ars.usda.gov

Emile Morse June 1999. *Evaluating Visual Information Browsing Displays*, Department of Information Science and Telecommunications, School of Information Sciences, University of Pittsburgh, Research Scientist, Information Technology Laboratory, National Institute of Standards and Technology, deceased

Jozsef Toth 1997. *The Effect of Representation in Computer-Mediated Small Group Decision Making*, Department of Information Science and Telecommunications, School of Information Sciences, University of Pittsburgh, previously Research Staff, Institute for Defense Analysis, deceased

Postdoctoral Researchers

Alin Coman (2010-2012) Associate Professor of Psychology, Princeton University, acoman@princeton.edu

Andreas Kolling (2009-2012). Principal Scientist, Amazon Robotics, Boston, MA andreas.kolling@gmail.com

Publications

Refereed Journals & Magazines

1. Lesong Jia, Kiran Shridhar Alva, Huao Li, Werner Hager, Lucia Romero, Timothy Albert Butler, and **Michael Lewis** (2026). Understanding Situation Awareness in Multi-UAV Supervision: Effects of Environmental Complexity,Interface Design,and Human/AI Differences, *International Journal of Human-Computer Interaction*.

2. Guo, Y.; Pang, Y.; Lyons, J.; **Lewis, M**.; Sycara, K.; Liu, R. Trust-Aware Reflective Control for Fault-Resilient Dynamic Task Response in Human–Swarm Cooperation. AI 2024, 5, 446-464.

3. Karten, S., Tucker, M., Li, H., Kailas, S., **Lewis, M**. & Sycara, K. (2023) Interpretable Learned Emergent Communication for Human-Agent Teams, IEEE Transactions on Cognitive and Developmental Systems

4. August Capiola, Izz aldin Hamdan, Elizabeth L. Fox, Joseph B. Lyons, Katia Sycara & **Michael Lewis** (2021) "Is something amiss?" Investigating individuals' competence in estimating swarm degradation, Theoretical Issues in Ergonomics Science, DOI: 10.1080/1463922X.2021.1983887

5. Li, H., Ni, T., Agrawal, S., Jia, F., Raja, S., Gui, Y., Hughes, D., **Lewis, M**., Sycara, K. (2021). Individualized Mutual Adaptation in Human-Agent Teams, *IEEE Transactions on Human-Machine Systems,* 51(6), 706-714.

6. Chandarana, M., Hughes, D., **Lewis, M**., Sycara, K. and Scherer, S. (2021). Planning and Monitoring Multi-Job Type Swarm Search and Service Missions, *Journal of Intelligent & Robotic Systems*, Vol. 101, No. 3, March, 2021

7. Joseph B. Lyons, J., Sycara, K., **Lewis, M.** & Capiola, A. (2021). Human-Autonomy Teaming: Definitions, Debates, and Directions, *Frontiers in Psychology:*

8. Chien, S., **Lewis, M.**, Sycara, K., Liu, J. & Kumru, A. (2020). Influence of Culture, Transparency, Trust, and Degree of Automation on Automation Use, *IEEE Transactions on Human-Machine Systems*, 50(3), 205-214.

9. Nam, C., Walker, P., Li, H., **Lewis, M.,** Sycara, K. (2020). Models of Trust in Human Control of Swarms with Varied Levels of Autonomy, *IEEE Transactions on Human-Machine Systems*, 50(3), 194-204.

10. Chien, S., **Lewis, M.**, Sycara, K., Liu, J. & Kumru, A. (2018). The Effect of Culture on Trust in Automation: Reliability and Workload, *ACM Transactions on. Interactive Intelligent Systems*. 8(4), article 29. ACM.

11. Chien, S., Lin, Y., Lee, P., Han, S., **Lewis, M.** & Sycara, K. (2018) Attention Allocation for Human Multi-Robot Control: Cognitive Analysis based on Behavior Data and Hidden States, *International Journal of Human-Computer Studies*. 117, 30-44.

12. Kolling, A., Walker, P., Chakraborty, N., Sycara, K., **Lewis, M**. (2015). Human Interaction with Robot Swarms: A Survey. In *Human-Machine Systems (HMS), 2015 IEEE Transactions on*. (pp. 9-26). IEEE.

13. Kolling, A., Sycara, K., Nunnally, S., & **Lewis, M**. (2013). Human swarm interaction: An experimental study of two types of interaction with foraging swarms, *Journal of Human-Robot Interaction*, 2(2), 103-128.

14. Kleiner, A., Kolling, A., **Lewis, M**., & Sycara, K. (2013) Hierarchical Visibility for Guaranteed Search in Large-Scale Outdoor Terrain*, Journal of Autonomous Agents and Multi-Agent Systems*, 26(1), 1-36.

15. **Lewis, M**., Wang, H., Chien, S., Ma, Z., Velagapudi, P., Scerri, P., & Sycara, K. (2011) Process and performance in human-robot teams, *Journal of Cognitive Engineering and Decision Making,* 5(2), 186-208.

16. Scerri, P., Ma, Z., Chien, S., Wang, H., Lee, P., **Lewis, M**. & Sycara, K. (2011) An initial evaluation of approaches to building entry for large robot teams, *Journal of Intelligent and Robotic Systems ,*64, 145-159.

17. **Lewis, M**., Wang, H., Chien, S., Velagapudi, P., Scerri, P. & Sycara, K. (2010). Choosing autonomy modes for multirobot search, *Human Factors*, 52(2), 225-233.

18. **Lewis, M**., Sycara, K., & Scerri, P. (2009). Scaling up wide-area-search-munition teams, *IEEE Intelligent Systems*, 24(3), 10-13.

19. Sycara, K., Glinton, R., Yu, B., Giampapa, J., Owens, S., **Lewis, M**., & Grindle, C. (2009). An integrated approach to high-level information fusion, *Information Fusion*, 10(1), 25-50.

20. Balakirsky, S., Carpin, S., Kleiner, A., **Lewis, M**., Visser, A., Wang, J., and Zipara, V. (2007). Toward hetereogeneous robot teams for disaster mitigation: Results and performance metrics from RoboCup Rescue, *Journal of Field Robotics*, 24(11-12), 943-967.

21. **Lewis, M**., Wang, J., and Hughes, S. (2007). USARsim : Simulation for the Study of Human-Robot Interaction, *Journal of Cognitive Engineering and Decision Making*, 1(1), 98-120.

22. **Lewis, M**. and Wang, J. (2007). Gravity referenced attitude display for mobile robots: Making sense of what we see, *Transactions on Systems, Man and Cybernetics Part A*, 37(1), 94-105.

23. Hughes, S. and **Lewis, M**. (2005). Partially guided viewpoint control in virtual environments, *Human Factors*, 47(3), 630-643.

24. Hughes, S. and **M. Lewis**. (2005). Task-Driven Camera Operations for Robotic Exploration. *Transactions on Systems, Man and Cybernetics Part A*, IEEE, 35(4), 513-522.

25. Jacobson, J. and **Lewis, M**. (2005). Game engine virtual reality with CaveUT. *IEEE Computer*, 38(4), 79-82.

26. Nourbakhsh, I., Sycara, K., Koes, M., Young, M., **Lewis, M**., and Burion, S. (2005). Human-Robot Teaming for Search and Rescue, *Pervasive Computing*, IEEE, 72-78.

27. Prasithsangaree, P., Manojlovich, J., Hughes, S., and **Lewis, M**. (2004). UTSAF: A Multi-Agent-Based Software Bridge for Interoperability between Distributed Military and Commercial Gaming Simulation, *Simulation*, 80(12), 647-657.

28. Giampapa, J., Sycara, K., Owens, S., Glinton, R., Seo, Y., Yu, B., Grindle, C., and **Lewis, M**. (2004). Extending the ONESAF Testbed into a C4ISR Testbed, *Simulation*, 80(12), 681-691.

29. **Lewis, M**. & Jacobson, J. (2002) Game Engines in Research. *Communications of the Association for Computing Machinery* (CACM), NY: ACM 45(1), 27-48.

30. Morse, E., **Lewis, M**., and Olsen, K. (2002) Testing visual information retrieval methodologies case study: comparative analysis of textual, icon graphical and 'spring' displays. *Journal of the American Society for Information Science and Technology* (JASIST), Silver Spring, MD: ASIST 53(1) 28-40

31. Morse, E., **Lewis, M**. and Olsen, K. (2000) Evaluating visualizations: Using a taxonomic guide. *International Journal of Human Computer Studies*, 53(5), 637-662.

32. **Lewis, M**. (1998). Designing for human-agent interaction. *AI Magazine.*67-78.

33. **Lewis, M**. (1993). Assessing decision heuristics using machine learning. *Decision Support Systems*, 10, 199-212.

34. **Lewis, M**. & Sycara, K. (1993). Reaching informed agreement in multi-specialist cooperation, *Group Decision and Negotiation*, vol 2, 3, 279-300.

35. **Lewis, Michael** (1992). Situated visualization: Building interfaces from the mind up. *Multimedia Review*, 4(1), Spring, 23-40.

36. **Lewis, M**. & Heidorn, B. (1991). Identifying tacit strategies in aircraft maneuvers. *IEEE Transactions on Systems, Man, and Cybernetics*, 21(6), 1560-1571.

37. Sycara, K. & **Lewis M**. (1991). Modeling Group Decision making and negotiation in concurrent product design. *Systems Automation: Research and Applications*, 1(3), 217-238

38. Weiss, M. & **Lewis, M**. (1991). Telecommunications pricing and consumer expectation: The case of alternate operator services. *Telecommunications Policy*, Dec 91, 497-508.

39. **Lewis, C. Michael** and Stine, Wm Wren (1989) Hidden dependence in human errors, *IEEE Transactions on Reliability*, 38:3, NY: IEEE, 296-300.

40. Rouse, William B., Hammer, John M., and **Lewis, C. Michael** (1989) Algorithmic approaches to knowledge engineering, *IEEE Transactions on Systems, Man, and Cybernetics*, 19, 558-573.

41. Jamison, Wesley and **Lewis, C. Michael** (1990) UnixTutor: A Menu-Based Transitional Interface, *SIGCHI Bulletin*, NY: ACM.

42. **Lewis, C. Michael** and Hammer, John M. (1986) Significance testing of rules in rule-based models of human problem solving. *IEEE Transactions on Systems, Man, and Cybernetics*, vol SMC-16, no. 1,154-157

43. Gerth, J.M., **Lewis, C.M**., Stine, Wm, Maple, T.I. (1982) Evaluation of two computerized data collection devices for research in zoos. *Zoo Biology*, New York: Alan R. Liss, Inc.

Book Chapters

1. **Lewis, Michael**, Li, Huao, Sycara, Katia (2020). "Deep Learning, transparency and trust in Human Robot Teamwork", In Trust in Human-Robot Interaction, Nam, Chang and Lyons, Joseph (eds), Elsevier.

2. Walker, P., Miller, C., Mueller, J., Sycara, K. and **Lewis, M**. (2019). A Playbook-based interface for human control of swarms, In M. Mouloua and P. Hancock (Ed.), Human Performance in Automated and Autonomous Systems, CRC Press, 61-88.

3. **Lewis, M**., Sycara, K. and Walker, P. (2017). The role of trust in human-robot interaction, In Abbass, H., Scholz, J., & Reid, D. (Eds.), Foundations *of Trusted Autonomy,* Springer-Verlag, Berlin-Heidelberg. 135-159.

4. Nunnally, S., Walker, P., **Lewis, M**., Chakraborty, N. and Sycara, K. (2015). Using haptic feedback in human swarm interaction, In Ying Tan (Ed.), *Handbook of Research on Design, Control, and Modeling of Swarm Robotics*, Hershey, PA: IGI Global.

5. **Lewis, M.** (2013). Human Interaction with Multiple Remote Robots, In D. Kaber (Ed), Human Factors Reviews Volume 9, on *Human Performance in Teleoperation and Beyond*, HFES, 131-174.

6. Sukthankar, Gita and Shumaker, Randall and **Lewis, Michael** (2011) Intelligent Agents as Teammates. In: Theories of Team Cognition: Cross-Disciplinary Perspectives. Routledge Academic, pp. 313-343.

7. **Lewis, M**. & Wang, J. (2010). Coordination and automation for controlling robot teams, in M. Barnes & F. Jentsch (Eds), Human-Robot Interactions in Future Military Operations, Burlington, VT: Ashgate, 397-418.

8. **Lewis, M**. and Wang, J. (2009). Assessing coordination demand in cooperating robots, In Madhavan, Tunstel and Messina (Ed.) Performance Evaluation and Benchmarking of Intelligent Systems, Springer, New York, 169-186.

9. Y. Xu, P. Scerri, **M. Lewis**, and K. Sycara (2008). Token-based approach for scalable team coordination," in *Cooperative Networks: Control and Optimization*, Edward Elgar Publishing.

10. Wang, J. and **Lewis, M**. (2007). Mixed-initiative multirobot control in USAR, In N. Sarkar Ed., *Human-Robot Interaction*, Itech Education and Publishing: Vienna, Austria, 405-422.

11. **Lewis, M**., Polvichai, J., Sycara, K., and Scerri, P. (2006) Scaling-up Human Control for Large UAV Teams, In N. Cooke (Ed.), *The Human Factors of Remotely Piloted Vehicles*, New York: Elsevier, pp. 237-250.

12. Wang, J. **Lewis, M**., and Hughes, S. (2006) USARsim: Simulation for the Study of Urban Search and Rescue Robotics, In S. Carpin, A. Birk, A. Jacoff, A. Nardi, and S. Tadokoro (Eds), *Rescue Robotics, Lecture Notes in Computer Science*, Springer-Verlag, Berlin Heidelberg, 98-120.

13. Xu, Y., Liao, E., Scerri, P., Yu, B., K. Sycara, and **Lewis, M**. (2006) Towards Flexible Coordination of Large Scale Multi-Agent Teams, in *Coordination of Large Scale Multiagent systems*, Springer, 287-209.

14. Scerri, P., Xu, Y., Polvichai, J., Yu, B., Okamoto, S., **Lewis, M**. and Sycara, K. (2005). Challenges in Building Very Large Teams, in *Massively Multiagent Systems I*, 86-103.

15. Scerri, P., Xu, Y., Liao, E., Lai, G., **Lewis, M**. and Sycara, K. (2004) Coordinating large groups of wide area search munitions. In D. Grundel, R. Murphey, and P. Pandalos (Ed.), *Recent Developments in Cooperative Control and Optimization*, Singapore: World Scientific, 451-480.

16. Sycara, K. and **Lewis, M**. (2004). Integrating intelligent agents into human teams. In E. Salas and S. Fiore (Eds.) *Team Cognition: Understanding the Factors that Drive Process and Performance*, Washington, DC: American Psychological Association, 203-232.

17. Toth, J. & **Lewis, M**. (2002) The Role of working memory and external representation in individual decision making In M. Anderson, B. Meyer, and P. Olivier (Eds.) *Diagrammatic Representation and Reasoning*, Springer- Verlag, Berlin-Heidelberg, 207-221.

18. **Lewis, M**. (1999). Anticipation, delegation, and demonstration: Why talking to agents is hard. In Klusch, Shehory, & Weiss (Eds): *Cooperative Information Agents III, Lecture Notes in Artificial Intelligence* vol. 1652, Springer-Verlag, Berlin Heidelberg, 365-389.

19. **Lewis, M**. and J. Toth (1994). Situated cognition in diagrammatic reasoning, AAAI *Technical Report on Reasoning with Diagrammatic Representations*, SS-92-02 Menlo Park, CA: AAAI. 47-52.

20. **Lewis, M**. & Sycara, K. (1993). Modeling multispecialist decision making, In M. Smith & G. Salvendy (Eds.) *Human-Computer Interaction: Applications and Case Studies*., New York: Elsevier, 481-486.

21. **Lewis, M**. (1991). Visualization and situations. In J. Barwise, M. Gawron, G. Plotkin, & S. Tutiya (Eds.), *Situation Theory and Its Applications, II*, Stanford, CA: CSLI Publications, 553-580.

Refereed Conferences and Workshops

1. Benjamin Li, Shuyang Shi, Lucia Romero, Huao Li, Yaqi Xie, Woojun Kim, Stefanos Nikolaidis, Michael Lewis, Katia P. Sycara, Simon Stepputtis (2025). Adaptively Coordinating with Novel Partners via Learned Latent Strategies, *Thirty-ninth Conference on Neural Information Processing Systems* (NeurIPS '25).

2. Benjamin Li, Shuyang Shi, Lucia Romero, Huao Li, Yaqi Xie, Woojun Kim, Stefanos Nikolaidis, Michael Lewis, Katia P. Sycara, Simon Stepputtis (2025). Modeling Latent Partner Strategies for Adaptive Zero-Shot Human-Agent Collaboration, *Robots, Science and Systems Workshop on Generative Models x HRI* **best paper award**

3. Huao Li, Hossein Nourkhiz Mahjoub, Behdad Chalaki, Vaishnav Tadiparthi, Kwonjoon Lee, Ehsan Moradi Pari, Michael Lewis, Katia Sycara (2024). Language Grounded Multi-Agent Communication for Ad-hoc Teamwork, *Thirty-eighth Conference on Neural Information Processing Systems* (NeurIPS '24).

4. Guo, X., Peng, A., Jia, L., Lewis, M. (2024) Target Conspicuity for Human-UAV Visual Perception, *IEEE International Conference on Human Machine Systems* (ICHMS)

5. Gia, L., Peng, A., Li, H., Guo, X., Lewis, M. (2024) Situation Theory Based Query Generation for Determining Situation Awareness Across Distributed Data Streams, *IEEE International Conference on Human Machine Systems* (ICHMS)

6. Guo, X., Li, H., Peng, A., Jia, L., Rishi, B., Lewis, M. (2024) Determining Human Perceptual Envelope in Fixed Wing UAV Surveillance*, IEEE International Conference on Human Machine Systems* (ICHMS)

7. Li, H., Chong, Y., Stepputtis, S., Campbell, J., Hughes, D., Lewis, M. & Sycara, K. (2023). Theory of Mind for Multi-Agent Collaboration via Large Language Models, *Empirical Methods in Natural Language Processing* (EMNLP '23).

8. Stepputtis, S., Campbell, J., Xie, Y., Qi, Z., Wang, R., Rangreji, S., Lewis, M. & Sycara, K. (2023). Long-Horizon Dialogue Understanding for Role Identification in the Game of Avalon with Large Language Models*, Empirical Methods in Natural Language Processing* (EMNLP '23).

9. Li, H., Fan, Y., Zheng, K., **Lewis, M.**, Sycara, K. (2023). Personalized Decision Supports based on Theory of Mind Modeling and Explainable Reinforcement Learning, *2023 IEEE Conference on Systems, Man, and Cybernetics.*

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11. Chis, M., Li, H., Zheng, K., **Lewis, M.**, Hughes, D., & Sycara, K. (2023). The Cognitive Load – Productivity Tradeoff in Task Switching, *Proceedings of the 67th Annual Meeting of the Human Factors and Ergonomics Society*.

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